# SURDS AND INDICES

# घातांक और करणी



## PRO LEVEL SHEET

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1. If A = $\sqrt[8]{\sqrt[3]{(81^2)^3}}$ and B = $\sqrt[6]{\sqrt[3]{(3^4)^3}} \times \sqrt[6]{\sqrt[6]{(3^2)^6}}$ then what will the value A + B?

यदि A = $\sqrt[8]{\sqrt[3]{(81^2)^3}}$ तथा B = $\sqrt[6]{\sqrt[3]{(3^4)^3}} \times \sqrt[6]{\sqrt[6]{(3^2)^6}}$ तो A + B का मान क्या है?

(a) 9 (b) 4
(c) 6 (d) 3

2. The greatest of the numbers $(2.89)^{0.5}$, $2-(0.5)^2$, $\left(1+\frac{0.5}{1-\frac{1}{2}}\right)$, $\sqrt{3}$ is

सबसे बड़ी संख्या $(2.89)^{0.5}$, $2-(0.5)^2$, $\left(1+\frac{0.5}{1-\frac{1}{2}}\right)$, $\sqrt{3}$ है

(a) $(2.89)^{0.5}$ (b) $2-(0.5)^2$
(c) $\left(1+\frac{0.5}{1-\frac{1}{2}}\right)$ (d) $\sqrt{3}$

3. The value of $\sqrt{11+2\sqrt{18}}$ is closest to/का मान किसके निकटतम है?

(a) 4.8 (b) 4.4
(c) 3.8 (d) 4.1

4. If $a^x = b$, $b^y = c$, $c^z = a$ and $b^2 = ac$, then y equals:

यदि $a^x = b$, $b^y = c$, $c^z = a$ और $b^2 = ac$ है, तो y बराबर है:

(a) $\frac{xy}{x+z}$ (b) $\frac{xz}{2(x-z)}$
(c) $\frac{xz}{2(z-x)}$ (d) $\frac{2xz}{(x+z)}$

5. The value of $\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}}}}}}$ is.

$\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}}}}}}$ का मान है

(a) 2 (b) $2^2$
(c) $2^3$ (d) $2^5$

6. Simplify/सरल करें :

$\left\{(6.25)^{\frac{1}{2}} \times (0.0144)^{\frac{1}{2}} + \{0.027\}^{\frac{1}{3}} \times (81)^{\frac{1}{4}}\right\}$

(a) 0.14 (b) 1.4
(c) 1 (d) 1.2

7. If $3^{2x+3} - 244 \times 3^x = -9$, then which of the following statements is true?

यदि $3^{2x+3} - 244 \times 3^x = -9$, तो निम्नलिखित में से कौन सा कथन सत्य है?

(a) 'X' is a positive number
(b) 'X' is a negative number
(c) 'X' can be either a positive number or a negative number
(d) None of the above

8. Given that $10^{0.48} = x$, $10^{0.70} = y$ and $x^z = y^2$, then the value of z is close to:

यह देखते हुए कि $10^{0.48} = x$, $10^{0.70} = y$ और $x^z = y^2$, तो z का मान किसके करीब है:

(a) 45 (b) 88
(c) 2.9 (d) 3.7

9. If $\sqrt{0.05 \times 0.5 \times a} = 0.5 \times 0.05 \times \sqrt{b}$, then $\frac{a}{b}$ is equal to

यदि $\sqrt{0.05 \times 0.5 \times a} = 0.5 \times 0.05 \times \sqrt{b}$ तो $\frac{a}{b}$ बराबर है

(a) 0.0025 (b) 0.025
(c) 0.25 (d) 0.00025

10. What is the value of :/का मान क्या है?

$\left(\sqrt{272-\sqrt{272-\sqrt{272-....\infty}}}\right) \div \left(\sqrt{20-\sqrt{20-\sqrt{20-....\infty}}}\right)$

(a) 4 (b) 2
(c) 3.6 (d) 8

11. If a, b, c are non-zero and $14^a = 36^b = 84^c$, then $6b\left(\frac{1}{c}-\frac{1}{a}\right)$ is equal to

यदि a, b, c शून्येतर हैं और $14^a = 36^b = 84^c$ है, तो $6b\left(\frac{1}{c}-\frac{1}{a}\right)$ बराबर है

(a) 3 (b) 2
(c) 1 (d) 5

12. The square root of/का वर्गमूल ज्ञात करें?

$\frac{0.75^3}{1-0.75} + [0.75 + (0.75)^2 + 1]$?

(a) 4 (b) 3
(c) 2 (d) 1

13. The value of $\frac{1}{4-\sqrt{15}} - \frac{1}{\sqrt{15}-\sqrt{14}} + \frac{1}{\sqrt{14}-\sqrt{13}} - \frac{1}{\sqrt{13}-\sqrt{12}} + \frac{1}{\sqrt{12}-\sqrt{11}} - \frac{1}{\sqrt{11}-\sqrt{10}} + \frac{1}{\sqrt{10}-3} - \frac{1}{3-\sqrt{8}}$ is:

$\frac{1}{4-\sqrt{15}} - \frac{1}{\sqrt{15}-\sqrt{14}} + \frac{1}{\sqrt{14}-\sqrt{13}} - \frac{1}{\sqrt{13}-\sqrt{12}} + \frac{1}{\sqrt{12}-\sqrt{11}} - \frac{1}{\sqrt{11}-\sqrt{10}} + \frac{1}{\sqrt{10}-3} - \frac{1}{3-\sqrt{8}}$ का मान ज्ञात करें।

(a) $2+2\sqrt{2}$ (b) $2-2\sqrt{2}$

(c) $4-2\sqrt{2}$ (d) $4+2\sqrt{2}$

14. If $3^x = 9^y = 27^z$ and

$\frac{1}{3x}+\frac{1}{6y}+\frac{1}{9z}=\frac{32}{3}$, find z.

यदि $3^x = 9^y = 27^z$ और $\frac{1}{3x}+\frac{1}{6y}+\frac{1}{9z}=\frac{32}{3}$ है, तो z ज्ञात कीजिए।

(a) $\frac{1}{5}$ (b) $\frac{5}{8}$

(c) $\frac{1}{32}$ (d) $\frac{8}{5}$

15. Given that $x^{2018}\, y^{2017} = \frac{1}{2}$ and $x^{2016}\, y^{2019} = 8$, the value of $x^2 + y^3$ is

यह देखते हुए कि $x^{2018}\, y^{2017} = \frac{1}{2}$ और $x^{2016}\, y^{2019} = 8$, $x^2 + y^3$ का मान है।

(a) $\frac{37}{4}$ (b) $\frac{31}{4}$

(c) $\frac{35}{4}$ (d) $\frac{33}{4}$

16. Find the value of $x$ in the following equation.

निम्न समीकरण में $x$ का मान ज्ञात करें।

$8^{2.3} \times 4^{0.8} \times 16^{0.4} = 2^x$

(a) 11.4 (b) 10.2

(c) 7.7 (d) 10.1

17. Given $A = 2^{65}$ and $B = (2^{64} + 2^{63} + 2^{62} + ...+2^0)$, which of the following is true?

$A = 2^{65}$ और $B = (2^{64} + 2^{63} + 2^{62} + ...+2^0)$ दिया गया है, निम्नलिखित में से कौन सा सत्य है?

(a) B is $2^{64}$ larger than A

(b) A and B are equal

(c) B is larger than A by 1

(d) A is larger than B by 1

18. On simplification

$$\frac{(49)^{\frac{1}{2}} \times (512)^{0.25} \times (8)^{0.25} \times (729)^{\frac{2}{3}} \div (64)^{\frac{4}{3}}}{(343)^{\frac{1}{3}} \times (256)^{-0.375} \times (81)^{0.5}}$$

reduces to:

$$\frac{(49)^{\frac{1}{2}} \times (512)^{0.25} \times (8)^{0.25} \times (729)^{\frac{2}{3}} \div (64)^{\frac{4}{3}}}{(343)^{\frac{1}{3}} \times (256)^{-0.375} \times (81)^{0.5}}$$

का न्यूनतम मान है-

(a) 2.25 (b) 1.25

(c) 1.5 (d) 2.5

19. If $9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$, then $x$ is

यदि $9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$, तो $x$ है

(a) $\frac{3}{2}$ (b) $\frac{2}{5}$

(c) $\frac{3}{4}$ (d) $\frac{4}{9}$

20. If $(5.55)^x = (0.555)^y = 1000$, then the value of $\frac{1}{x}-\frac{1}{y}$ is

यदि $(5.55)^x = (0.555)^y = 1000$, तो $\frac{1}{x}-\frac{1}{y}$ का मान है

(a) 1 (b) $\frac{1}{3}$

(c) $\frac{2}{3}$ (d) 3

21. If $2^{x+y-2z} = 8^{8z-5-y}$; $5^{4y-6z} = 25^{y+z}$; $3^{4x-3z} = 9^{x+z}$ then the value of $2x + 3y + 5z$ is:

यदि $2^{x+y-2z} = 8^{8z-5-y}$; $5^{4y-6z} = 25^{y+z}$; $3^{4x-3z} = 9^{x+z}$ है तो $2x + 3y + 5z$ का मान बताइए।

(a) 56 (b) 44

(c) 32 (d) 28

22. If $\left[\left\{\left(\frac{2}{3}\right)^3\right\}^{(2x+3)}\right]^{\frac{-3}{4}} = \left[\left\{\left(\frac{2}{3}\right)^{\frac{2}{3}}\right\}^{(3x+7)}\right]^{\frac{-6}{5}}$, then the value of $\sqrt{2-42x}$ is:

यदि $\left[\left\{\left(\frac{2}{3}\right)^3\right\}^{(2x+3)}\right]^{\frac{-3}{4}} = \left[\left\{\left(\frac{2}{3}\right)^{\frac{2}{3}}\right\}^{(3x+7)}\right]^{\frac{-6}{5}}$ है, तो

$\sqrt{2-42x}$ का मान ज्ञात कीजिए।

(a) 5 (b) 6

(c) 3 (d) 4

23. If cube root of 175616 is 56, then the value of $\sqrt[3]{175.616} + \sqrt[3]{0.175616} + \sqrt[3]{0.000175616}$ is equal to

यदि 175616 का घनमूल 56 है, तो $\sqrt[3]{175.616}$, $\sqrt[3]{0.175616}$, $\sqrt[3]{0.000175616}$ का मान बराबर है

(a) 0.168 (b) 62.16

(c) 6.216 (d) 6.116

24. If $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ and B

$= \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$, then what is A × B?

यदि $A = \frac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$ और B

$= \frac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}}$ है, तो A × B क्या है?

(a) $5 \times 10^{-8}$ (b) $7 \times 10^{-7}$

(c) $\left(\frac{7}{4}\right) \times 10^{-8}$ (d) $6 \times 10^{-8}$

25. If $x = 5 + 2\sqrt{6}$, then $\frac{(x-1)}{\sqrt{x}}$ is equal to:

यदि $x = 5 + 2\sqrt{6}$ है, तो $\frac{(x-1)}{\sqrt{x}}$ बराबर है:

(a) $\sqrt{2}$ (b) $2\sqrt{2}$

(c) $\sqrt{3}$ (d) $2\sqrt{3}$

26. Let $x = \sqrt{6-\sqrt{6+\sqrt{6-\sqrt{6+.....\infty}}}}$ then $x$ equals

मान लीजिए $x = \sqrt{6-\sqrt{6+\sqrt{6-\sqrt{6+.....\infty}}}}$ तो $x$ बराबर है

(a) $\frac{\sqrt{21}-1}{2}$ (b) $\frac{\sqrt{21}+1}{2}$

(c) 2 (d) 12

27. The value of / का मान ज्ञात कीजिए?

$$\frac{1}{\sqrt{3.25}+\sqrt{2.25}}+\frac{1}{\sqrt{4.25}+\sqrt{3.25}}+$$

$$\frac{1}{\sqrt{5.25}+\sqrt{4.25}}+\frac{1}{\sqrt{6.25}+\sqrt{5.25}}$$

(a) 1.00 (b) 1.25

(c) 1.50 (d) 2.25

28. If $\frac{\sqrt{26-7\sqrt{3}}}{\sqrt{14+5\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$ b > 0, then what is the value of $\sqrt{(b-a)}$ ?

यदि $\frac{\sqrt{26-7\sqrt{3}}}{\sqrt{14+5\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$ b > 0 हो, तो $\sqrt{(b-a)}$ का मान कितना होगा?

(a) 5 (b) 25

(c) 12 (d) 9

29. $\frac{1}{\sqrt{11-2\sqrt{30}}}-\frac{3}{\sqrt{7-2\sqrt{10}}}-\frac{4}{\sqrt{8+4\sqrt{3}}}$

(a) 0 (b) 1

(c) $\sqrt{2}$ (d) $\sqrt{5}$

30. If $\left(\frac{p^{-1}q^2}{p^3q^{-2}}\right)^{\frac{1}{3}} \div \left(\frac{p^6q^{-3}}{p^{-2}q^3}\right)^{\frac{1}{3}} = p^aq^b$, then the value of a + b, where P and q are different positive primes, is

यदि, $\left(\frac{p^{-1}q^2}{p^3q^{-2}}\right)^{\frac{1}{3}} \div \left(\frac{p^6q^{-3}}{p^{-2}q^3}\right)^{\frac{1}{3}} = p^aq^b$ है तो a + b का मान है, जहाँ P और q अलग-अलग धनात्मक अभाज्य संख्याएँ हैं

(a) –1 (b) $-\frac{2}{3}$

(c) $\frac{4}{3}$ (d) 2

## ANSWER KEY

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | (c) | 2. | (c) | 3. | (b) | 4. | (d) | 5. | (a) | 6. | (d) | 7. | (c) | 8. | (c) | 9. | (b) | 10. | (a) |
| 11. | (a) | 12. | (c) | 13. | (c) | 14. | (d) | 15. | (c) | 16. | (d) | 17. | (d) | 18. | (a) | 19. | (a) | 20. | (b) |
| 21. | (b) | 22. | (a) | 23. | (c) | 24. | (d) | 25. | (b) | 26. | (a) | 27. | (a) | 28. | (a) | 29. | (a) | 30. | (b) |

## SOLUTIONS

1. (c) $A = \sqrt[8]{\sqrt[3]{(81^2)^3}} = \sqrt[8]{\sqrt[3]{81^6}}$

$= \sqrt[8]{(81)^{6\times\frac{1}{3}}} = \sqrt[8]{81\times81} = 3^{8\times\frac{1}{8}} = 3$

$B = \sqrt[6]{\sqrt[3]{(3^4)^3}} \times \sqrt[6]{\sqrt[6]{(3^2)^6}}$

$= \sqrt[6]{\sqrt[3]{3^{12}}} \times \sqrt[6]{\sqrt[6]{3^{12}}}$

$= \sqrt[6]{3^{12\times\frac{1}{3}}} \times \sqrt[6]{3^{12\times\frac{1}{6}}} = \sqrt[6]{3^4} \times \sqrt[6]{3^2}$

$= 3^{4\times\frac{1}{6}} \times 3^{2\times\frac{1}{6}} = 3^{\frac{2}{3}} \times 3^{\frac{1}{3}} = 3^1 = 3$

$\therefore A + B = 3 + 3 = 6$

2. (c) $(2.89)^{\frac{1}{2}}, 2-0.25, \left(1+\frac{0.5}{1-\frac{1}{2}}\right), \sqrt{3}$

$\Rightarrow (1.7), 1.75, 2, 1.732$

$\Rightarrow$ Largest value $= \left(1+\frac{0.5}{1-\frac{1}{2}}\right)$

3. (b) $\sqrt{11+2\sqrt{18}} = \sqrt{\left(\sqrt{9}+\sqrt{2}\right)^2} = \sqrt{9}+\sqrt{2}$

$= 3 + 1.4 = 4.4$

4. (d) ATQ,

$a^x = b^y = c^z = k$ (Let)

$a = k^{\frac{1}{x}}, b = k^{\frac{1}{y}}, c = k^{\frac{1}{z}}$

$b^2 = ac$

$\Rightarrow k^{\frac{2}{y}} = k^{\frac{1}{x}} \times k^{\frac{1}{z}}$

$\Rightarrow \frac{2}{y} = \frac{1}{x} + \frac{1}{z}$

$\Rightarrow \frac{2}{y} = \frac{x+z}{xz}$

$\Rightarrow y = \frac{2xz}{x+z}$

5. (a) $\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}}}}}}....$

Let $x = \sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\sqrt[3]{4}}}}}}....$

Square both sides,

$\Rightarrow x^2 = 2\sqrt[3]{4x}$

Cubing both sides.

$\Rightarrow x^6 = 8 \times 4x \Rightarrow x^5 = 32$

$\Rightarrow x^5 = 2^5$

$\therefore x = 2$

6. (d) $\left\{(6.25)^{\frac{1}{2}} \times (0.0144)^{\frac{1}{2}} + (0.027)^{\frac{1}{3}} \times (81)^{\frac{1}{4}}\right\}$

$= 2.5 \times 0.12 + 0.3 \times 3$

$= 0.3 + 0.9 = 1.2$

7. (c) ATQ,

$3^x (3^x \times 27 - 244) = -9$

Let, $3^x = y$

$y(27y - 244) = -9$

$27y^2 - 244y + 9 = 0$

$y = 9, y = \frac{1}{27}$

$3^x = \frac{1}{27}$

$3^x = 3^{-3}$

$x = -3$

$3^x = 9$

$3^x = 3^2$

$x = 2$

$x$ is either negative or positive

8. (c) $x^z = y^2 \Rightarrow (10^{0.48})^z = (10^{0.70})^2$

$\Rightarrow 10^{(0.48)z} = 10^{1.40}$

$\Rightarrow 0.48z = 1.40$

$\Rightarrow Z = \frac{140}{48} = 2.9$

9. (b) $\sqrt{0.05 \times 0.5 \times a} = 0.5 \times 0.05 \times \sqrt{b}$

Squaring both sides

$0.05 \times 0.5 \times a = 0.5 \times 0.5 \times 0.05 \times 0.05 \times b$

$\frac{a}{b} = 0.025$

10. (a) $\left(\sqrt{272 - \sqrt{272 - \sqrt{272 - \sqrt{272 - \ldots\infty}}}}\right) \div$

$17 \times 16$

$\left(\sqrt{20 - \sqrt{20 - \sqrt{20 - \ldots.\infty}}}\right)$

$5 \times 4$

In such questions, factorise the number in gap of 1. Smaller value will be the answer.

$\Rightarrow 16 \div 4 = 4$

11. (a) $14^a = 36^b = 84^c \Rightarrow 14^a = 6^{2b} = 84^c = k$ (Let)

$14 = k^{\frac{1}{a}}, 6 = k^{\frac{1}{2b}}, 84 = k^{\frac{1}{c}}$

$\Rightarrow 14 \times 6 = 84$

$\Rightarrow k^{\frac{1}{a}} \times k^{\frac{1}{2b}} = k^{\frac{1}{c}}$

$\Rightarrow \frac{1}{a} + \frac{1}{2b} = \frac{1}{c}$

$\Rightarrow \left(\frac{1}{c} - \frac{1}{a}\right) = \frac{1}{2b}$

$\Rightarrow 6b \times \frac{1}{2b} = 3$

12. (c) $\frac{0.75^3}{1 - 0.75} + \left[(0.75) + (0.75)^2 + 1\right]$

$\Rightarrow \frac{\frac{27}{64}}{\frac{1}{4}} + \left[\frac{3}{4} + \frac{9}{16} + 1\right]$

$\Rightarrow \frac{27}{16} + \frac{3}{4} + \frac{9}{16} + 1 \Rightarrow \sqrt{\frac{64}{16}} = 2$

13. (c) $\frac{1}{4 - \sqrt{15}} - \frac{1}{\sqrt{15} - \sqrt{14}} + \frac{1}{\sqrt{14} - \sqrt{13}} - \frac{1}{\sqrt{13} - \sqrt{12}} + \frac{1}{\sqrt{12} - \sqrt{11}} - \frac{1}{\sqrt{11} - \sqrt{10}} + \frac{1}{\sqrt{10} - 3} - \frac{1}{3 - \sqrt{8}}$

$= 4 + \sqrt{15} - (\sqrt{15} + \sqrt{14}) + \sqrt{14} + \sqrt{13} - (\sqrt{13} + \sqrt{12}) + \sqrt{12} + \sqrt{11} - (\sqrt{11} + \sqrt{10}) + \sqrt{10} + 3 - (3 + \sqrt{8})$

$= 4 + \sqrt{15} - \sqrt{15} - \sqrt{14} + \sqrt{14} + \sqrt{13} - \sqrt{13} - \sqrt{12} + \sqrt{12} + \sqrt{11} - \sqrt{11} - \sqrt{10} + \sqrt{10} + 3 - 3 - \sqrt{8} = 4 - \sqrt{8} = 4 - 2\sqrt{2}$

14. (c) $3^x = 9^y = 27^z \Rightarrow 3^x = 3^{2y} = 3^{3z}$

$\Rightarrow x = 2y = 3z$

$\frac{1}{3x} + \frac{1}{6y} + \frac{1}{9z} = \frac{32}{3}$

$= \frac{1}{9z} + \frac{1}{9z} + \frac{1}{9z} = \frac{32}{3} = \frac{1}{3z} = \frac{32}{3}$

$z = \frac{1}{32}$

15. (d) $x^{2018}\ y^{2017} = \frac{1}{2}$ .....(i)

$x^{2016}\ y^{2019} = 8$ .....(ii)

(i) ÷ (ii)

$\Rightarrow \frac{x^2}{y^2} = \frac{1}{16}$

$\Rightarrow y = 4x$ Put in (i)

$x^{2018} \times 4^{2017} \times x^{2017} = \frac{1}{2}$

$x^{4035} = \left(\frac{1}{2}\right)^{4035}$

$x = \frac{1}{2}$

$y = 2$

$\Rightarrow x^2 + y^3 = \frac{1}{4} + 8 = \frac{33}{4}$

16. (d) $8^{2.3} \times 4^{0.8} \times 16^{0.4} = 2^x$

$\Rightarrow (2^3)^{2.3} \times (2^2)^{0.8} \times (2^4)^{0.4} = 2^x$

$\Rightarrow 2^{6.9} \times 2^{1.6} \times 2^{1.6} = 2^x$

$\Rightarrow 2^{6.9 + 1.6 + 1.6} = 2^x$

$\Rightarrow 2^{10.1} = 2^x \Rightarrow x = 10.1$

17. (d) $A = 2^{65}$

$B = 2^{64} + 2^{63} + 2^{62} + \ldots\ldots + 2^0$

$2^0 + 2^1 + 2^2 + \ldots\ldots\ldots + 2^{64}$

$= \frac{1(2^{65} - 1)}{(2 - 1)} = (2^{65} - 1)$

So, A is greater than B by 1

18. (a) $\dfrac{(49)^{\frac{1}{2}} \times (512)^{0.25} \times (8)^{0.25} \times (729)^{\frac{2}{3}} \div (64)^{\frac{4}{3}}}{(343)^{\frac{1}{3}} \times (256)^{-0.375} \times (81)^{0.5}}$

$= \dfrac{7 \times 8^{3\times\frac{1}{4}} \times 8^{\frac{1}{4}} \times 9^{3\times\frac{2}{3}} \div 4^{3\times\frac{4}{3}}}{7 \times (2^8)^{-0.375} \times 3^{4\times 0.5}}$

$= \dfrac{7\times 8\times 81 \div 256}{7 \times \frac{1}{2^{8\times\frac{3}{8}}} \times 9} = \dfrac{\frac{7\times 8\times 81}{256}}{7\times\frac{1}{8}\times 9} = \dfrac{7\times 8\times 81\times 8}{256\times 7\times 9}$

$= \dfrac{64\times 81}{9\times 256} = \dfrac{36}{16} = 2.25$

19. (a) $9^{x-\frac{1}{2}} - 2^{2x-2} = 4^x - 3^{2x-3}$

$\Rightarrow 3^{(2x-1)} - 2^{(2x-2)} = 2^{2x} - 3^{(2x-3)}$

$\Rightarrow 3^{(2x-1)} + 3^{(2x-3)} = 2^{2x} + 2^{(2x-2)}$

$= 3^{2x}\left[\dfrac{1}{3} + \dfrac{1}{27}\right] = 2^{2x}\left[1 + \dfrac{1}{4}\right]$

$\Rightarrow 3^{2x}\left[\dfrac{10}{27}\right] = \left[\dfrac{5}{4}\right]2^{2x}$

$= 3^{2x} \times 8 = 27 \times 2^{2x}$

$\Rightarrow \left(\dfrac{3}{2}\right)^{2x} = \left(\dfrac{3}{2}\right)^3 \Rightarrow x = \dfrac{3}{2}$

20. (b) Given that,
$(5.55)^x = 10^3$
$(0.555)^y = 10^3$
ATQ,

$= 5.55 = 10^{\frac{3}{x}}$ .....(1)

& $0.555 = 10^{\frac{3}{y}}$ .....(2)

By dividing (1) by (2)

$\dfrac{10^{\frac{3}{x}}}{10^{\frac{3}{y}}} = \dfrac{5.55}{0.555}$

$10^{\left(\frac{3}{x}-\frac{3}{y}\right)} = 10$

$\dfrac{3}{x} - \dfrac{3}{y} = 1$

$\dfrac{1}{x} - \dfrac{1}{y} = \dfrac{1}{3}$

21. (b) $2^{x+y-2z} = 8^{8z-5-y}$; $5^{4y-6z} = 25^{y+z}$ ; $3^{4x-3z} = 9^{x+z}$

$x + y - 2z = 24z - 15 - 3y$

$x + 4y - 26z = -15$ .....(i)

Now, $5^{4y-6z} = 25^{y+z}$

$4y - 6z = 2y + 2z$

$y = 4z$ .....(ii)

Now, $3^{4x-3z} = 9^{x+z}$

$4x - 3z = 2x + 2z$

$x = \dfrac{5z}{2}$ .....(iii)

The value of eqn. (ii) & (iii) in eqn, (i)

$\dfrac{5z}{2} + 16z - 26z = -15$

$z = 2$

Now, $2x + 3y + 5z = 2 \times \dfrac{5z}{2} + 3 \times 4z + 5z = 22z$

$= 22 \times 2 = 44$

22. (a) $\left[\left[\left(\dfrac{2}{3}\right)^3\right]^{(2x+3)}\right]^{\frac{-3}{4}} = \left[\left[\left(\dfrac{2}{3}\right)^{\frac{2}{3}}\right]^{(3x+7)}\right]^{\frac{-6}{5}}$

$3 \times (2x+3) \times \dfrac{-3}{4} = \dfrac{2}{3} \times (3x+7) \times \dfrac{-6}{5}$

$90x + 135 = 48x + 112$

$42x - 23$

$\sqrt{2 - 42x}$

$\sqrt{2 + 23} = 5$

23. (c) $\sqrt[3]{175.616} + \sqrt[3]{0.175616} + \sqrt[3]{0.000175616}$
$\Rightarrow 5.6 + 0.56 + 0.056$
$\Rightarrow 6.216$

24. (d) $A = \dfrac{\sqrt{0.0004} \times \sqrt[3]{0.000008}}{\sqrt[4]{16000} \times \sqrt[3]{125000} \times \sqrt[4]{810}}$

$= \dfrac{0.02\times 0.02}{2\times 5\times 3\times 100}$

$= \dfrac{4}{2\times 5\times 3\times 10^6} = \dfrac{2}{15} \times 10^{-6}$

$B = \dfrac{\sqrt[3]{0.729} \times \sqrt[4]{0.0016}}{\sqrt{0.16}} = \dfrac{0.9\times 0.2}{0.4} = \dfrac{9}{20}$

$\therefore A \times B = \dfrac{2}{15} \times 10^{-6} \times \dfrac{9}{20}$

$= \dfrac{18}{300} \times 10^{-6} = 6 \times 10^{-8}$

25. (b) $x = 5 + 2\sqrt{6}$

$\sqrt{x} = \left(\sqrt{3} + \sqrt{2}\right)$

$\Rightarrow \dfrac{x-1}{\sqrt{x}} = \dfrac{4+2\sqrt{6}}{\sqrt{3}+\sqrt{2}} \Rightarrow 2\sqrt{2}\dfrac{\left(\sqrt{2}+\sqrt{3}\right)}{\left(\sqrt{2}+\sqrt{3}\right)} = 2\sqrt{2}$

26. (a) $x = \sqrt{6 - \sqrt{6 + \sqrt{6.....\infty}}}$

$\Rightarrow \dfrac{\sqrt{4\times 6 - 3} - 1}{2}$

$= \dfrac{\sqrt{21} - 1}{2}$

27. (a)

$\dfrac{1}{\sqrt{3.25} + \sqrt{2.25}} + \dfrac{1}{\sqrt{4.25} + \sqrt{3.25}} + \dfrac{1}{\sqrt{5.25} + \sqrt{4.25}}$

$$+\frac{1}{\sqrt{6.25}+\sqrt{5.25}}$$

$\Rightarrow \sqrt{3.25}-\sqrt{2.25}+\sqrt{4.25}-\sqrt{3.25}+\sqrt{5.25}$

$-\sqrt{4.25}+\sqrt{6.25}-\sqrt{5.25}$

$= \sqrt{6.25}-\sqrt{2.25}$

$= (2.5 - 1.5) = 1.00$

28. (a) $\frac{\sqrt{26-7\sqrt{3}}}{\sqrt{14+5\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$

$\Rightarrow \frac{\sqrt{52-14\sqrt{3}}}{\sqrt{28+10\sqrt{3}}} = \frac{b+a\sqrt{3}}{11}$

$\Rightarrow \frac{\sqrt{(7-\sqrt{3})^2}}{\sqrt{(5+\sqrt{3})^2}} = \frac{b+a\sqrt{3}}{11}$

$\Rightarrow \frac{7-\sqrt{3}}{5+\sqrt{3}} = \frac{b+a\sqrt{3}}{11}$

$\Rightarrow \frac{(7-\sqrt{3})(5-\sqrt{3})}{(5+\sqrt{3})(5-\sqrt{3})} = \frac{b+a\sqrt{3}}{11}$

$\Rightarrow \frac{35-7\sqrt{3}-5\sqrt{3}+3}{22} = \frac{b+a\sqrt{3}}{11}$

$\Rightarrow \frac{38-12\sqrt{3}}{22} = \frac{b+a\sqrt{3}}{11}$

$\Rightarrow \frac{19-6\sqrt{3}}{11} = \frac{b+a\sqrt{3}}{11}$

$\Rightarrow b = 19, a = -6$

$\therefore \sqrt{b-a} = \sqrt{19-(-6)} = \sqrt{19+6}$

$= \sqrt{25} = 5$

29. (a) $\frac{1}{\sqrt{11-2\sqrt{30}}} - \frac{3}{\sqrt{7-2\sqrt{10}}} - \frac{4}{\sqrt{8+4\sqrt{3}}}$

$= \frac{1}{\sqrt{(\sqrt{6}-\sqrt{5})^2}} - \frac{3}{\sqrt{(\sqrt{5}-\sqrt{2})^2}} - \frac{4}{\sqrt{(\sqrt{6}+\sqrt{2})^2}}$

$= \sqrt{6}+\sqrt{5}-\left[\frac{3(\sqrt{5}+\sqrt{2})}{3}\right]-\left[\frac{4(\sqrt{6}-\sqrt{2})}{4}\right]$

$= \sqrt{6} + \sqrt{5} - \sqrt{5} - \sqrt{2} - \sqrt{6} + \sqrt{2} = 0$

30. (b) $\left(\frac{p^{-1}q^2}{p^3q^{-2}}\right)^{\frac{1}{3}} \div \left(\frac{p^6q^{-3}}{p^{-2}q^3}\right)^{\frac{1}{3}} = p^aq^b$

$\Rightarrow \left(\frac{q^4}{p^4}\right)^{\frac{1}{3}} \div \left(\frac{p^8}{q^6}\right)^{\frac{1}{3}}$

$\frac{q^{\frac{4}{3}}}{p^{\frac{4}{3}}} \times \frac{q^{\frac{6}{3}}}{p^{\frac{8}{3}}} = p^aq^b \Rightarrow \frac{q^{\frac{10}{3}}}{p^4} = p^aq^b$

$\Rightarrow q^{\frac{10}{3}}p^{-4} = p^aq^b$

$\Rightarrow a = -4 \quad b = \frac{10}{3}$

$a + b = -4 + \frac{10}{3} \Rightarrow -\frac{2}{3}$